

2

a



The label design is to create a binary response of roughly even split. Here the threshold is quality ≥ 6 , which results in good class balance, interpretability (roughly same amount of quality levels above and below threshold for both wines) and applied evenly to both datasets.

b

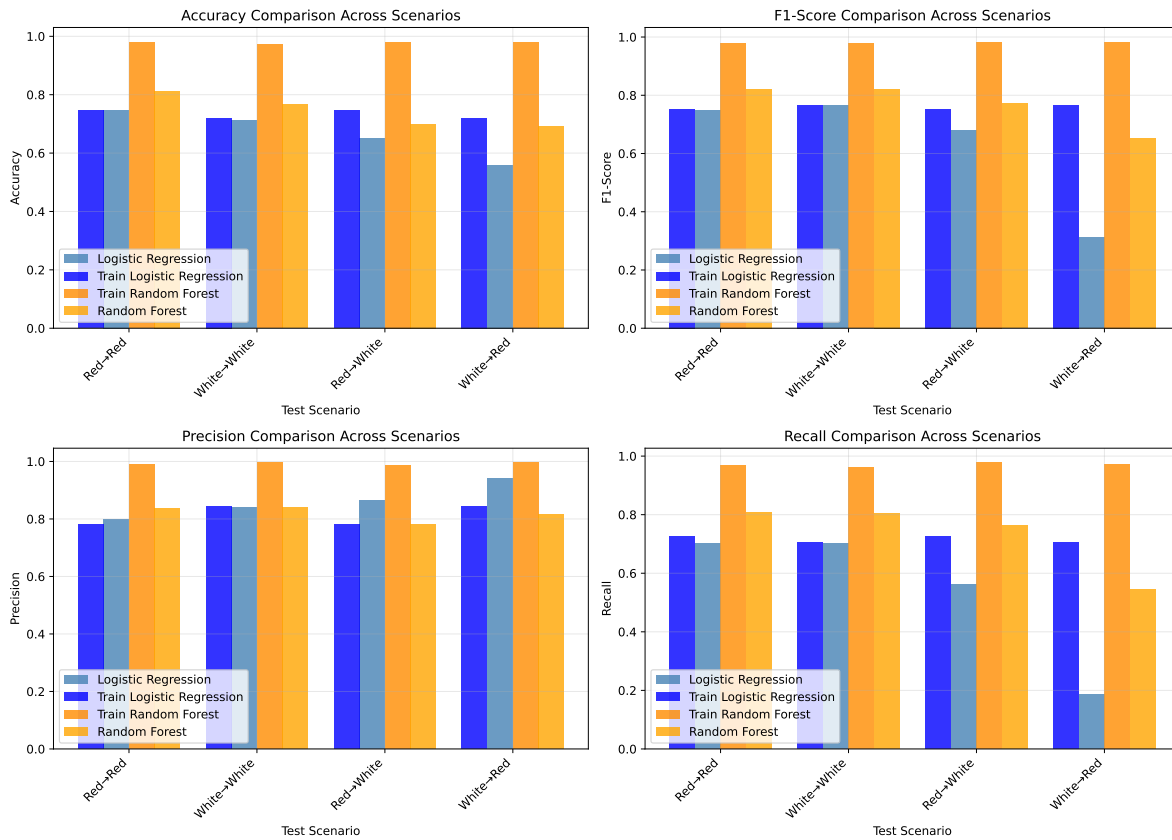
For this section, I used an 80/20 test/training split and the models I chose were Logistic and Random Forest. For the models, I used all of the features and did not perform any sort of feature selection. As mentioned in the problem, since the red wine set has significantly less data points than the white wine set, I decreased the white wine training set to match the size

of the red wine training set (1279 samples). The results of applying the models to the data is shown in the table below.

Testing Scenario	Model	Accuracy	Precision	Recall
Red → Red (In-domain)	Logistic Regression	0.747	0.800	0.702
Red → Red (In-domain)	Random Forest	0.812	0.836	0.807
White → White (In-domain)	Logistic Regression	0.712	0.839	0.702
White → White (In-domain)	Random Forest	0.767	0.836	0.804
Red → White (Cross)	Logistic Regression	0.650	0.864	0.563
Red → White (Cross)	Random Forest	0.699	0.779	0.764
White → Red (Cross)	Logistic Regression	0.559	0.941	0.187
White → Red (Cross)	Random Forest	0.691	0.816	0.544

C

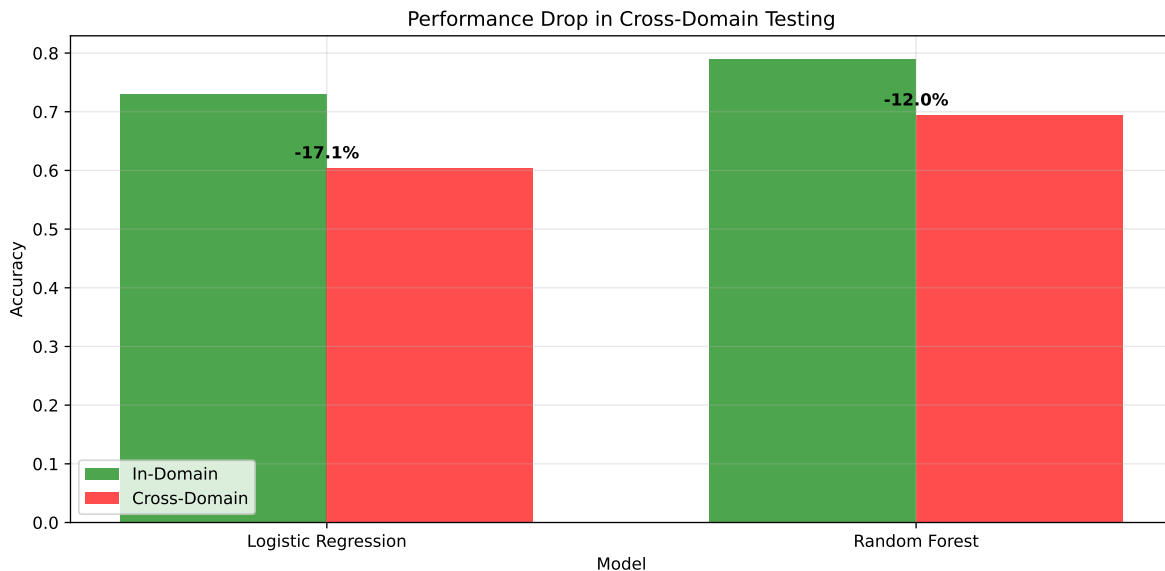
1) In Domain Model Consistency



When looking solely at the in domain scenarios within the graph, we see that Random Forest outperforms the linear regression for all metrics in all in domain scenarios (except for maybe precision in White -> White). However when looking at consistency, it appears the models are similar in consistency with a slight edge to the random forest. If we look at the accuracy graph, the difference between the accuracies in the in domain scenarios for logistic regression is roughly 0.035, while the accuracy difference for the Random Forest is roughly 0.045. For precision, Logistic Regressions had a difference of 0.039 while Random Forest was roughly 0. Lastly, for recall Logistic Regression had a difference of 0 and the Random Forest had a difference 0.03.

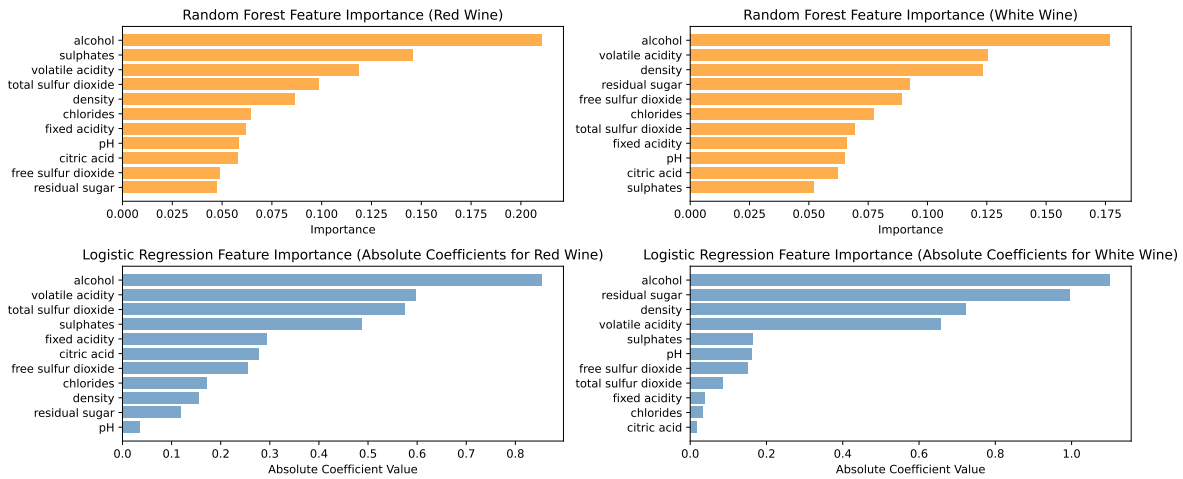
When looking at overfitting, it appears that the logistic regression is not overfitted as the different performance metrics increase and decrease by very small margins between training and testing. Random Forest however, there is significant decrease in all performance metrics when going from training to testing suggesting the presence of overfitting.

2) Cross Domain Model Generalization



Using the information from this graph, the table in 2b, and the graph in 2c1, it appears that the random forest performs slightly better than the logistic regression for cross domain. With the exception of precision, random forest outperforms the logistic regression model in every metric. The logistic regression decreased accuracy by an average of 0.13 whereas the random forest model decreased accuracy by an average of 0.1 when switching from in domain to cross domain.

3) Cross Domain Performance Differences



When trying to determine what factors that could explain the performance differences across domains, perhaps one of the biggest factors is the shift in feature importance across domains. From the graph we can see that both Logistic Regression and Random Forest models both think that the residual sugar is quite important for white wine predictions but not very important for red wine predictions. Essentially, the models do not perform well cross domain because it is weighing factors as important while the factor may not necessarily be important. A good analogy would be determining performance of football athletes based on age, weight, and height. Weight might be an important factor for linemen, but not as important for determining performance of a wide receiver.